

سبب

$$x_t \quad t \in \{0, \dots, T\}$$

Forward Process $q(x_t | x_{t-1}) = \mathcal{N}(\sqrt{1-\beta_t} x_{t-1}, \beta_t I)$

$$x_t = \sqrt{1-\beta_t} x_{t-1} + \sqrt{\beta_t} \epsilon \quad \epsilon \sim \mathcal{N}(0, I)$$

energy $\leftarrow \mathbb{E} \{ \|x_t\|^2 \} = (1-\beta_t) \mathbb{E} \{ \|x_{t-1}\|^2 \} + \beta_t \mathbb{E} \{ \|\epsilon\|^2 \} + 2\sqrt{1-\beta_t}\sqrt{\beta_t} \mathbb{E} \{ x_{t-1}^T \epsilon \}$

(by induct.)

= d

$$\mathbb{E} \{ \|\epsilon\|^2 \} = \mathbb{E} \{ \sum_{i=1}^d \epsilon_i^2 \} = \sum_{i=1}^d \mathbb{E} \{ \epsilon_i^2 \} = d$$

$$x_0 \rightarrow x_1 \rightarrow x_2 \rightarrow \dots \rightarrow x_T$$

Reverse Backward Process $x_T \xrightarrow{\text{denoising}} x_{T-1} \rightarrow x_{T-2} \rightarrow \dots \rightarrow x_0$

$\mathcal{N}(0, I)$

$$q(x_{t-1} | x_t) = ?$$

$$p_\theta(x_{t-1} | x_t)$$

Estimation Problem: Given $x_0^{(1)} \dots x_0^{(n)}$, estimate θ s.t.

$$p_\theta \approx q.$$

Evidence

$$\max_{\theta} \log p_\theta(x_0)$$

$$\max_{\theta} \sum_{i=1}^n \log p_\theta(x_0^{(i)})$$

marginal dist.

$$= \log \int p_\theta(x_0, x_1, \dots, x_T) \frac{q(x_1, \dots, x_T | x_0)}{q(x_1, \dots, x_T | x_0)} dx_1, \dots, dx_T$$

$$= \log \mathbb{E} \left\{ \frac{P_{\theta}(x_{0:T})}{q(x_{1:T}|x_0)} \right\}$$

$$\geq \mathbb{E}_{q(x_{1:T}|x_0)} \left\{ \log \underbrace{P_{\theta}(x_{0:T})}_{P_{\theta}(x_T) \cdot P_{\theta}(x_{T-1}|x_T) \cdot P_{\theta}(x_{T-2}|x_{T-1}, \cancel{x_T}) \cdot \dots \cdot P_{\theta}(x_0|x_1, \cancel{x_2}, \dots, \cancel{x_T})} / q(x_{1:T}|x_0) \right\}$$

Evidence Lower Bound
 $\leadsto$ Markovian Backward Process

$$q(x_{1:T}|x_0) = q(\cancel{x_T}|x_0) q(\cancel{x_{T-1}}|x_T, x_0) q(x_{T-2}|x_{T-1}, \cancel{x_T}, x_0) \dots q(x_1|x_2, \dots, \cancel{x_T}, x_0)$$

$\leadsto$ Markovian Backward Process

$$\text{ELBO} = \mathbb{E}_{q(x_{1:T}|x_0)} \left\{ \log \frac{P_{\theta}(x_T)}{q(x_T|x_0)} + \log \frac{P_{\theta}(x_{T-1}|x_T)}{q(x_{T-1}|x_T, x_0)} + \dots + \log \frac{P_{\theta}(x_1|x_2)}{q(x_1|x_2, x_0)} + \log \frac{P_{\theta}(x_0|x_1)}{q(x_0|x_1)} \right\}$$

$$\text{KL-div.} \quad D_{\text{KL}}(P \parallel q) = \int P(x) \log P(x)/q(x) dx$$

$$= \mathbb{E}_P \{ \log P/q \}$$

$$\text{ELBO} = - \left[D_{\text{KL}}(q(x_T|x_0) \parallel P_{\theta}(x_T)) + D_{\text{KL}}(q(x_{T-1}|x_T, x_0) \parallel P_{\theta}(x_{T-1}|x_T)) + \dots + D_{\text{KL}}(q(x_1|x_2, x_0) \parallel P_{\theta}(x_1|x_2)) \right] + \mathbb{E}_{q(x_{1:T}|x_0)} \{ \log P_{\theta}(x_0|x_1) \}$$

$$P_{\theta}(x_{t-1} | x_t) = \mathcal{N}(\mu(x_t, \theta), \dots)$$

← Gaussian
← Gaussian

$$q(x_{t-1} | x_t, x_0) = \frac{q(x_t | x_{t-1}, \cancel{x_0}) \cdot q(x_{t-1} | x_0)}{q(x_t | x_0)}$$

$$\begin{cases} x_t = \sqrt{1 - \beta_t} x_{t-1} + \sqrt{\beta_t} \epsilon_t \\ x_{t-1} = \sqrt{1 - \beta_{t-1}} x_{t-2} + \sqrt{\beta_{t-1}} \epsilon_{t-1} \end{cases}$$

$$x_t = \underbrace{\sqrt{1 - \beta_t}}_{\alpha_t} \left[\sqrt{1 - \beta_{t-1}} x_{t-2} + \sqrt{\beta_{t-1}} \epsilon_{t-1} \right] + \sqrt{\beta_t} \epsilon_t$$

$$= \underbrace{\sqrt{\alpha_t \alpha_{t-1}}}_{\bar{\alpha}_t} x_{t-2} + \underbrace{\sqrt{\alpha_t} \sqrt{\beta_{t-1}} \epsilon_{t-1} + \sqrt{\beta_t} \epsilon_t}_{\sim \mathcal{N}(0, \alpha_t \beta_{t-1} + \beta_t)}$$

$$\begin{aligned} & 1 - \beta_t \\ &= \beta_{t-1} - \beta_t \beta_{t-1} + \beta_t \\ &\stackrel{\text{claim}}{=} 1 - \alpha_t \alpha_{t-1} \\ &= 1 - (1 - \beta_t)(1 - \beta_{t-1}) \end{aligned}$$

$$\underline{x_t} = \sqrt{\alpha_t \alpha_{t-1} \dots \alpha_0} \underline{x_0} + \sqrt{1 - \alpha_t \alpha_{t-1} \dots \alpha_0} \epsilon$$